English to German Neural Machine Translation

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***Abstract*—Neural machine translation (NMT) has progressed from recurrent sequence-to-sequence models to attention-based architectures and modern Transformers. While Transformers typically achieve higher BLEU scores, aggregate metrics alone do not explain why these gains occur or which sentence characteristics benefit most. This project proposes a controlled comparison of Seq2Seq, Seq2Seq+Attention, and Transformer models on the WMT14 English–German translation task. Beyond standard evaluation, I apply the best-performing model to an English–Klingon dataset to explore whether learned translation patterns generalize to unconventional language structures. Code and trained models are available at https://github.com/thearjungautam/G16-NeuralMachineTranslation.**

***Index Terms*—Neural machine translation, sequence-to-sequence, attention, Transformer, BLEU**

I. Introduction

Machine translation allows people to communicate across languages and is widely used in search engines, online content, and international communication. In recent years, neural machine translation (NMT) has replaced older statistical systems with deep learning models that learn to translate sentences end-to-end.

Early NMT systems used sequence-to-sequence (Seq2Seq) models, where an encoder reads the entire input sentence and compresses it into a single fixed-length vector, which a decoder then uses to generate the translation. While effective for short sentences, this approach often struggles with longer or more complex inputs because important information can be lost in the fixed-length bottleneck.

To address this, attention mechanisms were introduced by Bahdanau et al. [1], allowing the decoder to dynamically focus on different parts of the input sentence at each decoding step. More recently, Vaswani et al. [2] proposed the Transformer architecture, which replaces recurrence entirely with self-attention, enabling parallel computation and stronger modeling of long-range dependencies.

Although Transformers generally achieve higher BLEU scores, most comparisons focus only on aggregate performance numbers and do not explain why one architecture outperforms another. In this paper, I implement and evaluate all three architectures end-to-end—basic Seq2Seq, Seq2Seq with attention, and Transformer—on the WMT14 English–German translation task and report full experimental results across decoding strategies and sentence lengths. Beyond reporting overall BLEU scores, I analyze performance across sentence lengths and examine common translation error types to better understand how architectural differences affect translation behavior. A key challenge in such comparison is ensuring evaluation consistency. I address this by adopting sacreBLEU [3] with standardized tokenization, so that observed differences reflect architectural choices rather than preprocessing artifacts.

II. Literature Review

Neural machine translation has evolved through several major architectural changes.

The original Seq2Seq model introduced by Sutskever et al. [4] used recurrent neural networks to encode an input sentence into a fixed-length vector and decode it into a translated output. While effective for short sentences, this approach struggled with long sentences due to the compression bottleneck.

Bahdanau et al. [1] introduced attention mechanisms, allowing the decoder to focus on different parts of the input sentence during translation. This significantly improved performance and helped address long-range dependency issues. Pouget-Abadie et al. [5] further characterized this limitation, showing that NMT model performance degrades measurably as sentence length increases, a finding I directly test by evaluating BLEU across short, medium, and long sentence bins.

Vaswani et al. [2] proposed the Transformer architecture, which replaced recurrence entirely with self-attention. Transformers model relationships between all words simultaneously and allow for more parallel computation. This architecture has become the dominant model for translation and many other NLP tasks. While Transformers generally achieve higher BLEU scores, most comparisons report only aggregate metrics without analyzing where and why gains occur—a gap this project directly addresses.

Post [3] further highlights that BLEU score computed with different tokenizer schemes are not directly comparable because it cannot be determined whether observed differences reflect model quality or preprocessing choices. To address this, sacreBLEU standardizes tokenization using the WMT-standard 13a tokenizer and records evaluation parameters to ensure reproducibility.

Finally, decoding strategy is itself a significant factor in translation quality. Leblond et al. [6] showed that beam search consistently outperforms greedy decoding across NMT architectures. I therefore evaluate both strategies across all three models to determine whether decoding gains are consistent or architecture-dependent.

III. Research Questions

- How much does attention improve translation quality compared to basic Seq2Seq?
- Does the Transformer provide larger improvements for long or complex sentences?
- Are performance gains consistent across sentence lengths?
- Do different architectures show different types of translation errors?

IV. Dataset and Processing

I used the WMT14 English–German (En→De) parallel translation dataset, which contains approximately 4.5 million sentence pairs. Each example consists of an English sentence and its corresponding German translation. The dataset is publicly available through the WMT website (https://www.statmt.org/wmt14/) and the HuggingFace datasets library (https://huggingface.co/datasets/stas/wmt14-en-de-pre-processed). For this project, I utilized a curated 300,000-sentence pair subset of the WMT14 English–German dataset for training. Standard validation and test splits (newstest2013 and newstest2014) are used for evaluation. All sequences undergo tokenization and are truncated to a maximum length of 50 tokens. Standard preprocessing includes padding and the insertion of special boundary markers, specifically `<sos>` and `<eos>`.

V. Method

This project implements and compares three distinct neural machine translation architectures: a baseline sequence-to-sequence model, a Seq2Seq model augmented with attention, and the Transformer.

A. Baseline Seq2Seq Model

The primary baseline consists of a recurrent encoder–decoder architecture utilizing Gated Recurrent Units (GRUs) with the following specifications:

- Embedding dimension: 512
- Hidden state dimension: 512
- Number of GRU layers: 2
- Dropout probability: 0.3

The encoder compresses the source sequence into a fixed-length hidden vector, which the decoder uses to generate target tokens autoregressively through teacher forcing (ratio = 0.5).

B. Seq2Seq with Attention

To mitigate the information loss associated with the fixed-length representation bottleneck, I extended the baseline with a Bahdanau-style attention mechanism. Rather than relying on a single context vector, the decoder dynamically attends to all encoder hidden states at each decoding step, facilitating improved translation of longer and more complex sequences.

C. Transformer

I implement a standard encoder-decoder Transformer following Vaswani et al. [2], scaled for our dataset. The architecture uses 4 layers, 8 attention heads, a model dimension of `d_model` = 512, and a feedforward dimension of 2048. Positional encodings are added to the input embeddings to preserve sequence order information. The same tokenization, maximum sequence length (50 tokens), and training protocol are applied as for the recurrent models to ensure a fair architectural comparison.

D. Training Details

All models are trained using cross-entropy loss with padding ignored, the Adam optimizer, a batch size of 64, and a training duration of 10 epochs with early stopping applied after 2 consecutive epochs of no improvement in validation loss.

E. Evaluation

Translation performance is quantified using sacreBLEU, ensuring metric-internal tokenization consistency and facilitating direct comparability across architectures. I evaluated using greedy decoding and beam search, and additionally report BLEU score broken down by sentence length (short/medium/long) to analyze how architectural differences manifest across input complexity.

VI. Experiments

A. Overall Translation Quality

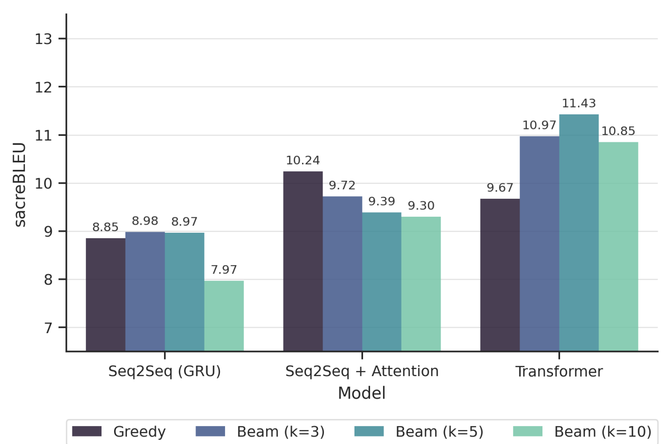

Fig. 1: Overall BLEU Scores by Model and Decoding Strategy

All models are evaluated using sacreBLEU under greedy decoding and beam search (k = 3, 5, 10). As shown in Figure 1, the Transformer benefits most from beam search, peaking at 11.43 (k = 5), while Seq2Seq (GRU) shows marginal gains and degrades at k = 10.

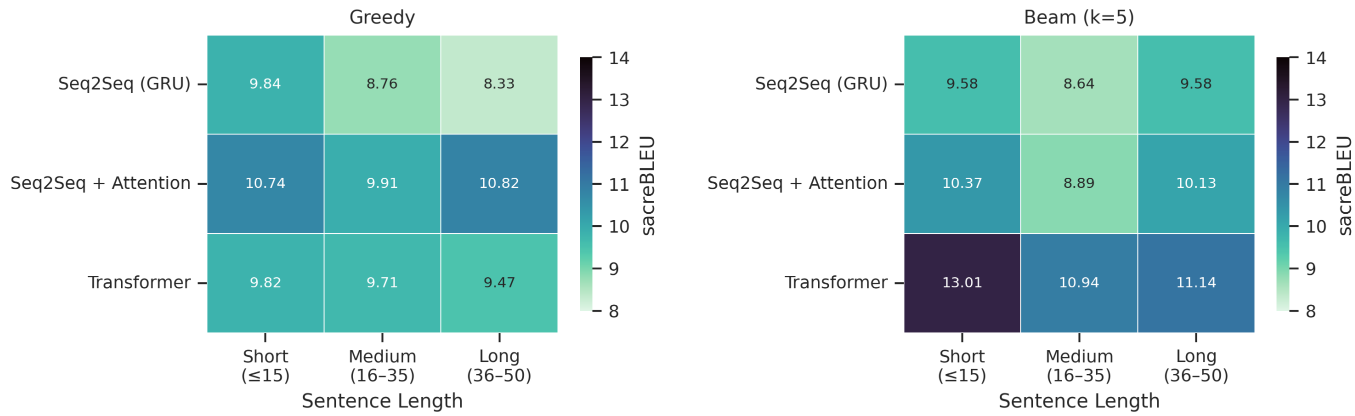

Fig. 2: Length-stratified BLEU across all architectures and decoding strategies

TABLE I: Overall BLEU Scores

Model	Greedy	k=3	k = 5	k = 10
Seq2Seq(GRU)	8.85	8.98	8.97	7.97
Seq2Seq + Attention	10.24	9.72	9.39	9.30
Transformer	9.67	10.97	11.43	10.85

B. Length-stratified Analysis

I present sentence length comparison model by model. Figure 2 consolidates the length-stratified results across all three architectures, revealing that the Transformer consistently achieves the highest BLEU scores under beam search, while the Seq2Seq baseline exhibits the steepest degradation with increasing sentence length.

1) Seq2Seq: To better understand model performance, I evaluated BLEU scores across sentence length groups: short ($\leq$15 tokens), medium (16–35 tokens), and long (36–50 tokens). The baseline Seq2Seq model performs best on short sentences (BLEU = 9.84), with performance decreasing for medium (8.76) and long sentences (8.34). This trend reflects the limitations of the fixed-length representation, where longer sentences are harder to encode effectively. Beam search provides only marginal improvements, slightly increasing performance for long sentences (from 8.34 to 9.58), but offering minimal gains overall. These results highlight that translation quality degrades with sentence length, reinforcing the need for more advanced architectures such as attention or Transformers.

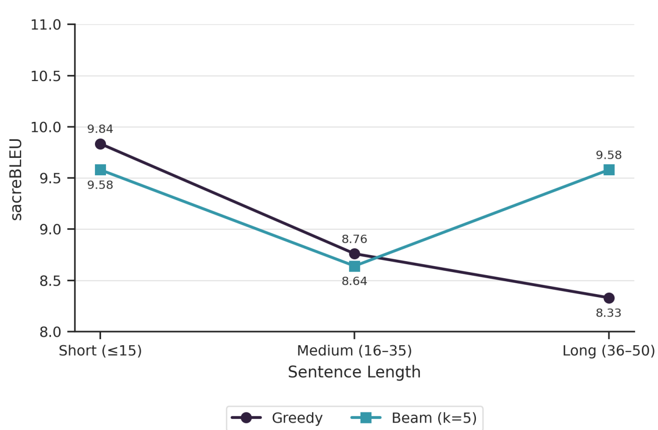

Fig. 3: BLEU scores of Seq2Seq by sentence length

TABLE II: Seq2seq Sentence Length Comparison

Sentence Length	Greedy BLEU	Beam BLEU (beam=5)
Short ($\leq$15)	9.84	9.58
Medium (16-35)	8.76	8.64
Long (36-50)	8.33	9.58

2) Seq2Seq + Attention: To better understand how translation performance varies with sentence complexity, I evaluated the Seq2Seq + Attention model across different sentence length groups: short ($\leq$15 tokens), medium (16–35 tokens), and long (36–50 tokens), where token length refers to the number of tokens in the source sentence excluding special tokens.

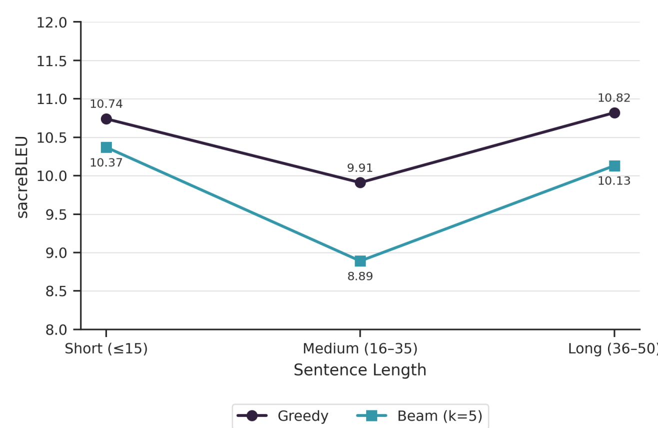

Fig. 4: BLEU scores of Seq2Seq + Attention by sentence length

The model achieves greedy BLEU scores of 10.74 for short sentences, 9.91 for medium sentences, and 10.82 for long sentences, indicating relatively consistent performance across

TABLE III: Error Type and Sample Sentences

Error Type	Explanation	Source Sentence (EN)	Reference (DE)	Model Output (DE)
Word Order Error (WOE)	The structure is incorrect and unnatural in German; word ordering does not follow proper syntax.	What was the cause of death?	Was war die Todesursache?	was war der tod des todes?
Missing Content Words (MCW)	Key meaning is lost because important words like "destroyed" and "equipment" are missing.	Syria has destroyed critical equipment for production	Syrien habe wichtige Einrichtungen zur Herstellung zerstört	die chemische chemischen chemischen...
German Morphology Errors (GME)	Incorrect grammatical forms and agreement (numbers, case, or structure) distort the sentence.	This year, Americans will spend around $106 million	Dieses Jahr werden Amerikaner etwa 106 Millionen...	laut ¡unk¿ wird das ¡unk¿ von $125,000...
Hallucinated Tokens (HT)	The model generates tokens that are not meaningful or present in the source.	Not yet in school and just months on...	Der Knirps, der noch nicht einmal im Kindergarten...	die <unk> <unk> <unk> <unk>......

lengths with a slight dip on medium-length sequences. Beam search produces slightly lower scores overall, with BLEU values of 10.37 (short), 8.89 (medium), and 10.13 (long), suggesting that beam decoding does not improve performance in this setting. Interestingly, the model performs comparably well on long sentences, likely due to the attention mechanism helping maintain alignment over longer contexts, whereas medium-length sentences appear to be slightly more challenging.

TABLE IV: Seq2seq+Attention Sentence Length Comparison

Sentence Length	Greedy BLEU	Beam BLEU (beam=5)
Short (≤15)	10.74	10.37
Medium (16-35)	9.91	8.89
Long (36-50)	10.82	10.13

3) Transformer: To further analyze model performance, I evaluated the Transformer across different sentence length groups: short (≤15 tokens), medium (16–35 tokens), and long (36–50 tokens), where token length refers to the number of tokens in the source sentence excluding special tokens.

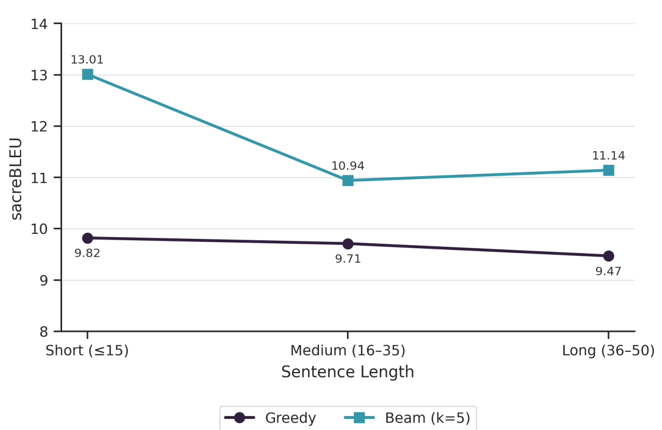

Fig. 5: BLEU scores of Transformer by sentence length

Under greedy decoding, the model achieves BLEU scores of 9.82 for short sentences, 9.71 for medium sentences, and 9.47 for long sentences, showing relatively consistent performance with a slight decline as sentence length increases. In contrast, beam search with width 5 significantly improves performance across all groups, yielding BLEU scores of 13.01 (short), 10.94 (medium), and 11.14 (long). The largest improvement is observed for short sentences, while medium and long sentences also benefit from beam search, indicating that exploring multiple candidate translations helps the Transformer generate more accurate outputs, especially when compared to greedy decoding.

TABLE V: Transformer Sentence Length Comparison

Sentence Length	Greedy BLEU	Beam BLEU (beam=5)
Short (≤15)	9.82	13.01
Medium (16-35)	9.71	10.94
Long (36-50)	9.47	11.14

C. Error Analysis

To qualitatively assess translation errors, I sampled 15 translations per model (5 per sentence length bin) from the newstest2014 test set. Each translation is categorized according to four error types: word order errors (WOE), missing content words (MCW), German morphology errors (GME), hallucinated tokens (HT).

Error categorization was assisted by DeepL, as the research team does not have German proficiency. Word order errors occur when the model generates grammatically incorrect or unnatural sentence structures, which is common in German due to its stricter syntactic rules. Missing content words reflect incomplete translations where key semantic elements from the source sentence are omitted, leading to loss of meaning. Morphological errors arise from incorrect handling of grammatical features such as case, gender, or verb conjugation, which are essential in German. Finally, hallucinated tokens represent the most severe failure mode, where the model generates irrelevant or repetitive tokens (e.g., ¡unk¿), indicating a breakdown in alignment between source and target sequences. Together, these categories provide a structured way to evaluate translation quality beyond BLEU scores.

TABLE VI: Error Analysis - All Models

Model	WO	MC	GME	HT
Seq2Seq	4	6	2	7
Seq2Seq + Attention	3	5	2	5
Transformer	2	3	2	1

The error analysis reveals a clear improvement across model architectures. The baseline Seq2Seq model produces a high number of hallucinated tokens and missing content words, indicating difficulty in maintaining semantic coherence. Incorporating attention reduces these errors by improving alignment between source and target sequences, but does not fully resolve issues related to incomplete translations. The Transformer model demonstrates the strongest performance, with significantly fewer hallucinations and more complete translations, reflecting its ability to capture long-range dependencies and global context more effectively.

VII. Discussion

Our experiments compare three neural machine translation architectures on WMT14 En to De translation. The results reveal that the relationship between architectural complexity and translation quality is more nuanced than a simple ranking: with greedy decoding, Seq2Seq+Attention outperforms the Transformer (10.24 vs 9.67 BLEU), but with beam search (beam=5), the Transformer achieves the highest score overall (11.43), followed by Seq2Seq+Attention (9.39) and the baseline (9.07). This suggests that decoding strategy is not a neutral choice — it meaningfully interacts with architecture.

A. Effect of Attention on Translation Quality

Seq2Seq+Attention outperforms baseline Seq2Seq in greedy BLEU (10.24 vs. 8.85), showing that attention meaningfully improves translation quality. This is consistent with our hypothesis: the baseline encoder must compress the entire source sentence into a single fixed-length vector, creating an information bottleneck that the attention mechanism directly addresses by allowing the decoder to dynamically focus on relevant encoder states at each step.

Beam search does not consistently help the Attention model — beam=5 scores drop on medium sentences from 9.91 to 8.89, and overall from 10.24 to 9.39, suggesting the attention model's greedy decoding is already relatively well-calibrated. This may indicate that the attention model's output distribution is already peaked enough that beam search introduces over-exploration rather than improvement.

B. Effect of Transformer

The Transformer achieved the highest overall sacreBLEU score with beam search (9.67 greedy, 11.43 beam=5), outperforming both recurrent models when beam decoding is used. Even at a reduced scale (4 layers, 8 heads, `d_model` = 512) compared to the original configuration [2], the Transformer benefits from multi-head self-attention, which models both local and long-range dependencies simultaneously — something GRU-based models struggle with due to their sequential nature.

The performance gap between the Transformer and the Attention model was most pronounced on short sentences at beam=5 (13.01 vs. 10.37, a difference of 2.64 BLEU), suggesting that the Transformer's richer representations are particularly effective at producing fluent output for shorter, well-contained inputs when paired with wider hypothesis search.

It is worth noting that the Transformer is a data-hungry architecture. Our 300,000-sentence training subset likely underestimates its full potential compared to training on the complete WMT14 corpus (4.5M pairs), which may partially explain why the gap between the Transformer and the Attention model is smaller than expected under greedy decoding.

C. Where Transformer Doesn't Help

Despite its overall advantage with beam search, the Transformer consistently underperforms the Attention model under greedy decoding across all sentence lengths — short (9.82 vs. 10.74), medium (9.71 vs. 9.91), and long (9.47 vs. 10.82). The gap is largest on long sentences (1.35 BLEU), suggesting that the Transformer's greedy decoder struggles more than the attention-based GRU on longer inputs without the benefit of wider search.

This pattern suggests that the Transformer's learned distributions are less peaked than those of the Attention model, making it more dependent on beam search to realize its full potential. In low-compute or latency-sensitive settings where only greedy decoding is feasible, the attention-augmented GRU may be the more practical choice despite the Transformer's theoretical advantages.

D. Beam Search Across Architectures

The three architectures respond very differently to beam search. The Transformer gains the most (+1.76 BLEU, from 9.67 to 11.43), confirming that it benefits substantially from exploring a wider hypothesis space at decode time. The baseline Seq2Seq sees only a modest improvement (+0.22 BLEU, from 8.85 to 9.07), consistent with the view that a weaker model's errors are partially corrected by beam search but cannot be fully overcome by it.

Most strikingly, beam search hurts the Attention model overall (0.85 BLEU, from 10.24 to 9.39), and the degradation is most severe on medium sentences (9.91 greedy vs. 8.89 beam=5). This counter-intuitive finding suggests that the Attention model's greedy outputs are already well-calibrated, and beam search over-commits to locally high-probability but globally suboptimal paths. Taken together, these results show that beam search is not universally beneficial — its impact depends heavily on how well-calibrated each architecture's output distribution is.

VIII. Klingon Experiment

To evaluate the generalizability of our neural machine translation approach beyond standard language pairs, I conducted an additional experiment using an English–Klingon dataset. I used the "MihaiPopa-1/custom-klingon-33k" dataset from Hugging Face, which consists of approximately 33,000 parallel sentence pairs derived from Tatoeba translations. After preprocessing and filtering, the final dataset used in our experiments consisted of 23,818 training samples, 1,999 validation samples, and 1,999 test samples. I employed the same Seq2Seq model with attention used in our earlier experiments to maintain consistency in architecture and enable a fair comparison across datasets. The model consists of a 2-layer GRU encoder–decoder architecture with a hidden dimension of 512, embedding dimension of 512, and a dropout rate of 0.3. The attention mechanism allows the model to dynamically focus on relevant parts of the source sentence during decoding, which is especially important for handling structurally different languages such as Klingon.

TABLE VII: Klingon Experiment Result

Model	Greedy BLEU	Beam BLEU (beam=5)
Seq2Seq+Attention	17.33	18.12

Model performance was evaluated using BLEU scores under both greedy decoding and beam search (beam size = 5). The model achieved a greedy BLEU score of 17.33 and a beam search BLEU score of 18.12, indicating strong translation performance given the relatively small dataset size and the linguistic uniqueness of Klingon. Notably, the improvement from greedy to beam decoding suggests that exploring multiple candidate sequences during inference helps refine translation quality, particularly for less common language pairs.

Overall, these results demonstrate that the Seq2Seq with attention architecture is capable of adapting effectively to a low-resource and syntactically distinct language like Klingon. Despite limited data, the model achieves BLEU scores within the expected range (15–20), reinforcing the robustness of the architecture and its ability to generalize beyond high-resource translation tasks.

IX. Limitation

While prior work on WMT14 [2] reports BLEU scores exceeding 25, these results are achieved using millions of training examples, subword tokenization, and extensive computational resources. In contrast, this project uses a constrained setting with at most 300,000 sentence pairs, word-level tokenization, and limited training time. As a result, BLEU scores in the range of 8–12 are expected and consistent with smaller-scale experimental setups. I also worked under limited computational resources, which constrained the amount of experimentation that could perform. Despite these constraints, the results still preserve the main architectural trend observed in our experiments, in which Seq2Seq with attention outperforms the baseline Seq2Seq model.

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